

BHOOMIKA RAVISHANKAR

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EDUCATION

University of Illinois Urbana-Champaign, Master of Science in Information Management, GPA: 4.0 /4.0

August 2024 - May 2026

Phi Kappa Phi Honor Society (Top 10% of Graduate Students)

Savitribai Phule Pune University, Bachelor of Engineering in Computer Engineering, GPA: 9.12 /10.0

August 2020 - June 2024

WORK EXPERIENCE

Synchrony Financial, Champaign, IL

May 2025 - May 2026

Data Scientist, Credit Risk Division

- Developed and deployed statistical **PD/EAD models** to production via AWS within Synchrony's AtLAS platform, a complementary log-log GLM framework predicting **5-year** charge-off probability across **6** retail portfolios (clothing, lawn & garden, lifestyle, luxury, sporting goods) directly performing **data quality checks** for **\$7.49B CECL loss reserve**.
- Engineered **13+ longitudinal features** spanning credit bureau scores, utilization groups, and behavioral interactions alongside macroeconomic signals (unemployment lag, GDP, savings rate, HPI, median income) across **15+ years** of snapshots; retained variables drove **75%** deviance reduction from the null model with **near-zero p-values** (hypothesis test) and stable **VIF ≤ 3** across interaction terms.
- Validated model robustness through multi-snapshot backtesting from **2007–2025**, confirming conservative under-prediction (**~10%**) pre-2008 for regulatory compliance, self-correction within **~1 year** following the 2008 financial shock, and stable charge-off variance across **S1–S4 stress** periods including post-COVID behavioral shifts.
- Built large-scale **PySpark ETL pipelines** on AWS EMR, scheduled via Apache Airflow, to ingest and transform **900M+** customer records per snapshot, reducing pipeline runtime from **1 hr (Python)** to **5 mins (PySpark)**, **92%** faster.

Data Scientist, Operations Analytics Division

- Replaced an external consulting firm's NPS model with an internally built solution, **eliminating third-party vendor costs**; findings on top NPS-driving survey questions were adopted by **customer analytics leadership** to redesign survey structure and cut low-signal questions.
- Migrated legacy **SAS/SQL workflows to Python**, performed **SQL performance tuning**, building a scalable ETL pipeline unifying **100M+** account-level records across 11 survey datasets, cutting runtime from **2 hr to 15 mins (87% faster)** and resolving schema mismatches & deduplication issues that blocked modeling.
- Modeled NPS drivers at the Apply Experience Survey level using XGBoost (**R²/RMSE similar, confirming no overfitting**) and **Lasso regression (R² 0.43)** in parallel, with mean/median imputation across **100K+** missing values; Lasso coefficients served as a cross-validation signal to identify which survey questions most consistently predicted NPS movement.
- Validated causal robustness using **Placebo Refutation Testing**, replacing the treatment variable with a random placebo and confirming **near-zero effect (-0.001)**, establishing that identified survey drivers reflect genuine signal rather than random correlation.

University of Illinois Urbana-Champaign, Champaign, IL

January 2025 - Present

Research Assistant, AI Engineer

- Built an end-to-end **LLM-powered data pipeline** in **Python** to convert **100,000 unstructured clinical trial studies** into structured model-ready variables using Qwen 2.5 7B-Instruct, entity extraction, prompt engineering, one-hot encoding, sentence embeddings, and PCA.
- Developed reusable **data transformation and structured-output workflows** to standardize noisy clinical text into validated downstream features, improving reliability for LLM-assisted modeling and research analysis.
- Benchmarked **LLM-derived features** against sentence embeddings using Lasso logistic regression, feature-importance analysis, and model evaluation, finding structured LLM features contributed **97.2% of total feature importance** versus 2.8% from embeddings.
- Led iterative **prompt refinement and evaluation cycles** using 0.79 F1 and 0.59 5-fold ROC-AUC as feedback signals to identify weak fields, redesign extraction prompts, rerun inference, and improve output quality.

Business Intelligence Group (BIG), Champaign, IL

January 2025 - May 2025

Data Consultant

- Designed and shipped a **RAG-based internal decision-support tool** using Python, retrieval-augmented generation, topic modeling, and Power BI dashboards to automate contract/query resolution, resource allocation, and overtime planning, reducing pilot lookup time by **90%**.
- Translated ambiguous business workflow needs into a working **AI-powered automation tool**, iterating with users to improve retrieval quality, dashboard usability, and adoption for operational decision-making.

PROJECTS

- AI-driven M&A Prediction**: Architected a multi-modal M&A target prediction system combining 14 LSEG financial ratios with **298K+ financial news headlines**, using **NER to extract company** and ticker mentions, **FinBERT sentiment**, **Word2Vec** across **3,882 companies**; applied **SMOTE to address 20:1 class imbalance** and **XGBoost with 5-fold CV**, achieving **89% accuracy** and **2.4× F1 improvement**.
- LSTM Credit Spend Forecasting**: Built a 2-layer LSTM (64→32) to forecast Q4 2025 credit card spend across 14,099 accounts using 13-month transaction histories, achieving \$2,444 MAE and enabling credit line recommendations for 288 high-value accounts (~2% segment).
- SQL Personal Finance Management System**: Designed and queried a normalized **MySQL relational database** for a personal finance system, reducing manual financial reporting from **~4 hours to under 30 seconds/month**; delivered **7 analytical reports** (budget-vs-actual variance, net worth VIEW, spending trends) and **11 CRUD operations** across 6 tables, using MySQL 8.0, aggregate functions, **CASE WHEN** logic, and **JOINS**.

TECHNICAL SKILLS

Programming & Quant Analytics: Python, SQL, R, SAS, PySpark, Spark, Pandas, NumPy, Scikit-learn, Git, Conda, Tableau, Power BI

Finance & Risk Modeling: Credit Risk, Loss Forecasting, PD/EAD, CECL, Allowance Modeling, CCAR, Stress Testing, Portfolio Risk

Management, Model Governance, Model Validation, Model Monitoring, Time-Series Forecasting, Backtesting, Hypothesis Testing

Statistics & Machine Learning: Causal Inference, Linear Regression, Logistic Regression, GLM, Lasso/Ridge, XGBoost, Random Forest

Cloud & Big Data: AWS, AWS S3, AWS EMR, AWS Glue, AWS SageMaker, Airflow, Databricks, Snowflake, BigQuery, Hadoop, Hive